Neuro-Symbolic Causal Inference via Language Model Knowledge Extraction and Bayesian Networks

Phase 1: Classical Machine Learning
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Abstract

This Phase 1 project separates text-based causal reasoning, directed relation classification, reviewed graph construction and Bayesian intervention inference. On a premise-disjoint Corr2Cause evaluation, classical TF-IDF with Logistic Regression obtains 0.700 macro F1, but a hypothesis-only control reaches 0.653, exposing shortcut sensitivity. A separate external SemEval-2010 Task 8 experiment evaluates directed causal relations with supplied entity spans. Context TF-IDF with tuned Logistic Regression reaches 0.820 directed edge F1 against 0.742 for lexical rules and 0.503 for an entity-only ablation on 2,700 retained test pairs. Approved edges with provenance can now define a Bayesian network fitted to matching binary observations. Synthetic intervention queries agree with known generating-process answers within 0.011. These results establish an auditable classical baseline while preserving the distinction between text claims, structural validity and causal identification.

Problem and scope

The project investigates how a language-facing component can propose causal structure while a formal graph component enforces structural constraints and computes explicit queries. The title supplied by the course is Neuro-Symbolic Causal Inference via LLM Knowledge Extraction and Bayesian Networks. For this submission, only the Advanced Machine Learning component is implemented. The language-facing model is TF-IDF with classical classifiers; there is no language-model service or neural training. The system has a Streamlit interface and a reproducible notebook. Its outputs distinguish reasoning labels, text mentions, reviewed graph edges, and probabilities from observational data.

Literature and methodological choices

Pearl describes causal models and interventions as changes to mechanisms, rather than ordinary conditioning [1]. This distinction motivates deleting incoming edges during an intervention. Jin et al. introduce Corr2Cause to test inference from stated statistical relations to causal conclusions [2]. Its synthetic examples provide a controlled reasoning task, but do not provide annotated scientific sentences for open-domain cause-effect extraction. We therefore do not interpret Corr2Cause classification scores as extraction scores. Scikit-learn documents sparse TF-IDF representations for text learning [3]. They provide a transparent, computationally inexpensive baseline whose feature weights can be inspected. pgmpy supplies discrete Bayesian models, parameter estimation and exact inference [4]. Together, these sources motivate separate evaluation of language classification, graph validation and causal queries. BERT is part of the broader project brief but is not implemented or evaluated in Phase 1.

Dataset, split audit and EDA

The source is the official causal-nlp/corr2cause dataset on Hugging Face [5], pinned to revision 42ba12c769e11ff6427c9f52d7db58e3f9bf3e53. To keep the local run bounded, we use the first 12,000 training rows and the complete development and test CSVs. This prefix is not a random sample and may underrepresent the full dataset. Every downloaded slice has a SHA-256 hash in data/raw/manifest.json. A row combines a premise and a hypothesis with a binary entailment label. Before fitting any text features, we normalize and hash the premise, retain the original training split, and remove premise groups already seen in earlier splits. This leaves 12,000 training rows, 1,040 development rows and 1,044 test rows. There are 213, 21 and 18 exact premise groups respectively. Thirty-six development rows and 118 test rows are removed. Exact input duplicates are also removed. The grouping does not guarantee separation of semantically equivalent graphs under variable renaming. The test set contains 888 negatives and 156 positives. This imbalance is why accuracy is not the selection criterion. Saved EDA includes class counts, word-length distributions, missing-input counts and relation-template slices. The additional extraction dataset is SemEval-2010 Task 8 [6], pinned to revision d58325907b9bf84a7c81f81fe5372791d911c10c. It supplies two marked nominal spans per sentence and 19 original relation labels. We map Cause-Effect(e1,e2) to class 1, Cause-Effect(e2,e1) to class 2, and all remaining relations to class 0. The original 8,000-row train partition yields 6,400 train and 1,600 dev examples through a seeded stratified group fold. Removing 17 normalized-sentence overlaps from original test leaves 2,700 rows, including 327 causal pairs. Grouping uses text without entity tags, while duplicate pair validation uses the marked sentence. Different pairs in one sentence may legitimately have different labels. Corpus document IDs are unavailable, so sentence disjointness does not guarantee document disjointness. This is a general semantic-relation extension, not validation on scientific/news text.

Features and classical models

The main representation is word TF-IDF with unigrams and bigrams, a maximum vocabulary of 18,000 features and sublinear term frequency for logistic regression. Single-letter tokens are retained because A, B and C denote variables in this dataset. The vectorizer is fitted inside a pipeline using training text only.

Smooth inverse document frequency is log((1 + N)/(1 + df)) + 1, followed by L2 normalization. The score is a sigmoid applied to a linear combination of sparse features. Logistic regression uses C = 1, balanced class weights, a maximum of 1,500 iterations and random seed 42. The alternatives are a most-frequent classifier and multinomial naive Bayes with alpha = 1. A unigram-only logistic model is a prespecified ablation, excluded from selection. Candidate models are selected by development macro F1. Test results are computed after selection and are never used to choose the winner. The reasoning baseline uses fixed settings. The separate extraction search uses development labels only. Positive-class precision, recall and F1 accompany accuracy and macro F1. Classifier probabilities are uncalibrated model scores, not causal confidence. In extraction, the context model substitutes entity strings with ENTITY1 and ENTITY2. A train-fitted unigram/bigram TF-IDF vocabulary has at most 30,000 features. Balanced Logistic Regression searches C in {0.1, 1, 10}. Development directed edge F1 selects the model, with smaller C breaking ties. The decision uses class argmax, without test-tuned thresholds. A lexical baseline detects explicit forward and reverse cues between the given spans and abstains on modal/negated sentences. A separate entity-only logistic model uses the two nominal strings without surrounding context as an ablation.

Measured reasoning results

The selected TF-IDF plus logistic regression model scores 0.661 development macro F1 and 0.700 test macro F1. Test positive-class precision is 0.392, recall is 0.923 and F1 is 0.551. Accuracy is 0.775. The confusion matrix, with actual labels as rows and predicted labels as columns, is [[665, 223], [12, 144]]. The majority classifier reaches 0.851 accuracy but zero positive-class recall, illustrating the weakness of accuracy on this split. Naive Bayes achieves 0.465 test macro F1. The unigram ablation achieves 0.521, so retaining local variable relationships substantially improves this bounded experiment, while independent validation remains necessary. A seed-42 group bootstrap, resampling the 18 test premise groups with replacement 300 times, yields an approximate 95% interval of [0.670, 0.734] for the selected macro F1. This interval describes this retained test sample; it is not a population guarantee. The model produces 223 false positives and 12 false negatives. Feature-weight and template-error exports support inspection of lexical shortcuts rather than claims that the model learned formal d-separation.

Text extraction and graph review

The extractor supports simple affirmative clauses of the form X causes Y, X leads to Y and X results in Y. A small alias map unifies rain/rainfall, sprinkler/sprinklers and wet grass. Each candidate retains its document identifier, sentence identifier, original sentence and extraction method. Sentences containing explicit uncertainty or negation markers are conservatively skipped. The parser is deliberately narrow: passive constructions, richer syntax, context-dependent entity resolution and implicit mechanisms are unsupported. The interface requires explicit approval for every proposed edge. The graph builder rejects self loops, duplicate edges and edges that would create a directed cycle. It uses source order, so the retained graph is not an optimal or uniquely identified causal graph. Sixteen authored regression documents contain 13 gold directed mentions. The extractor recovers ten and produces no additional mentions: precision 1.000, recall 0.769 and F1 0.870. This is a development regression fixture, not an independent extraction benchmark. A separate four-mention graph fixture retains three correct DAG edges and rejects one cycle-closing edge. Its perfect edge precision and recall establish expected behavior only on that fixture. An external given-entity evaluation now supplements this fixture, while scientific/news domain validation and automatic entity identification remain uncompleted.

Directed extraction evaluation

Directed edge scoring counts a correct nonzero label as a true positive. A wrong direction contributes one false positive and one false negative. On the 2,700 retained test pairs, context LR yields 268 true positives, 59 false positives and 59 false negatives: directed precision, recall and F1 are each 0.8196. Its three-class macro F1 is 0.8757. Lexical rules yield precision 0.8471, recall 0.6606 and directed F1 0.7423. The entity-only ablation yields precision 0.5324, recall 0.4771 and F1 0.5032. Always predicting no relation yields zero directed F1 despite high noncausal-class prevalence. Development selected C=10 with edge F1 0.7392. The sentence-group bootstrap 95% interval for frozen test edge F1 is [0.7865, 0.8506]. The result therefore supports context sensitivity relative to the chosen controls, not a proof of formal causal comprehension. Test rows were inspected only after development selection. A sentence-group bootstrap interval conditions on the frozen model and does not capture training or selection uncertainty. Saved predictions include every test row for metric reconstruction.

Bayesian model and interventions

The probability demonstration uses binary Rain (R), Sprinkler (S) and Wet grass (W), with edges R to S, R to W and S to W. These are declared assumptions for a synthetic system, not discoveries from text. The factorization is P(R,S,W) = P(R) P(S|R) P(W|R,S). We generate 6,000 observations with seed 42: P(R=1)=0.3; P(S=1|R=1)=0.05 and P(S=1|R=0)=0.6. Wet-grass probabilities for (R,S)=(0,0),(0,1),(1,0),(1,1) are 0.02, 0.8, 0.9 and 0.99. The strong association between rain and sprinkler use creates confounding. Conditional probability tables are fitted with a BDeu prior of equivalent sample size five. Variable elimination computes queries. For an intervention, the network removes incoming edges to the intervention variable and conditions on its chosen state. The remaining factor for that fixed variable cancels in normalization. For S=0, the observed wet-grass probability is 0.452 and the fitted interventional value is 0.275, against an oracle value of 0.284. For S=1, the corresponding values are 0.801, 0.846 and 0.857. The oracle uses P(W=1|do(S=s)) = sum_r P(W=1|r,s)P(r). Causal interpretation requires the declared graph, observed confounders, positivity and stable mechanisms. The new reviewed-network fitting module accepts only explicitly approved, provenance-bearing edges and matching complete binary observation columns. It rejects unresolved invalid edges, missing observations, nonbinary values and incomplete assumptions. The reviewer records a causal rationale and assumptions about observed confounders and stable mechanisms. The app can fit this graph and answer observational/interventional queries. A synthetic-data shortcut is explicitly labeled. Text frequencies and classifier probabilities never substitute for observed-variable data. The demonstration still cannot establish

that an uploaded real-world graph is correctly identified. Population intervention queries are supported; individual counterfactual inference is not claimed.

Reproducibility and software validation

The original reasoning experiment and diagnostics were rerun from the pinned source slices. The new extraction runner preserves exact train/dev/test membership, predictions, development tuning traces, fitted model, environment versions, source snapshots and SHA-256 hashes in a dedicated run directory. A separate verifier checks hashes, split isolation, development-only selection, prediction alignment, saved-model prediction reproduction and all test metrics. Local dataset slices and pickle estimators are generated rather than required for a fresh clone. Portable JSON linear-model snapshots enable offline application inference. The complete automated suite covers extraction direction errors, malformed spans, conflicting pair labels, portable inference parity, reviewed graph fitting, invalid observation contracts, oracle agreement, split isolation and application rendering. A GitHub Actions workflow runs tests on Python 3.12. The notebook includes an extension walkthrough. A five-member contribution record requires the team to enter actual work and supporting evidence; no implementation ownership is invented.

Discussion and conclusion

The external extraction comparison strengthens the language component without treating entailment as extraction. Context LR improves directed edge F1 by about 0.077 absolute over lexical rules and about 0.316 over the entity-only ablation. These are conditional results on one derived three-class task with supplied entity spans, not official SemEval leaderboard scores. The reasoning shortcut control and low positive precision constrain claims of causal understanding. Sentence grouping cannot eliminate unknown document overlap or semantic template similarity. Acyclicity and reviewer approval do not establish causal truth. The fitted graph bridge makes the system demonstrable end to end on a controlled teaching example, but real-domain claims need independent scientific/news annotations, measured variables, defensible graph structure, positivity and confounder assessment. Individual counterfactual inference and compact transformer comparisons remain future Phase 2 work, outside this requested Phase 1 submission.

References

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